Phase 4.25: Rank-4 Factor Fresh-Admission Convergence

Tactix development report for Andrew

27 August 2026

1 Outcome

The Rank-4 gap is reduced from 1.9850x to 1.3888x relative to current instrumented Kissat by repairing one portable Factor lifecycle mismatch. The accepted predecessor took a stable median 15.340 seconds, the candidate takes 10.733 seconds, and current Kissat takes 7.728 seconds. All three arms have under-0.66% spread over three paired node018 waves. The candidate is about 30.0% faster than its predecessor.

This is a search-quality recovery rather than a micro-optimization. The Factor rule still takes about 30 ms, but restoring one missing accepted grid reduces conflicts from 983,053 to 647,074, close to Kissat's 651,764. Exact tracing establishes identical heap order, candidate order, accepted grids, ticks, and early learned state after the repair.

The complete B28+C5 correctness gate and the one Claude-owned fuzz campaign both pass. No correctness repair was needed. The increment is therefore ready for promotion, after which Rank 32 is the next declared witness.

2 First divergent common state

The immutable Rank-4 input is `tierD24-26/15456f0d13b26f0b4b21525ef7d0d711-clique_n5_k10.sanitized.cnf.xz`, SHA-256 `1066d755...02e1`, expected UNSAT. The stable predecessor campaign has three Tactix times of 15.340, 15.440, and 15.340 seconds versus three identical 7.728-second Kissat runs. All six proofs verify.

The two Factor implementations accept exactly the same first 284 grids. At the next heap frontier, both pop external literal 521 at the same position, with score 5 and the same surrounding heap state. Kissat admits it and applies grid `[521,-31,-32]`; Tactix discards it because its local per-pass `armed` mirror is zero. Values, retirement state, persistent score, and heap order are otherwise equal.

Source tracing identifies the missing lifecycle transition. Adding the divider and quotient clauses marks both polarities of Kissat's new fresh variable. Tactix re-armed the pre-existing factors and remainders, but did not copy those two fresh clause-add marks into the local pass image. This is the first demonstrated worse-side owner; it is not a tick, heap, or candidate-order difference.

3 Portable repair and semantic parity

After an accepted composite grid adds its one fresh variable, Tactix now mirrors the already-authoritative marks for its positive and negative literals into the local `armed` image. It does not push either literal merely because it was created. A later touched-candidate update controls scheduling, exactly as in Kissat. No benchmark identity, rank, deadline, or tuned constant is present, and the proof-rule/strategy boundary and N1–N4 remain unchanged.

The frozen candidate and exact-tick campaigns establish parity for:

- 1,251 Factor heap pops and 1,309 viable seeds;
- 381 accepted grids and all 1,971 ordered grid members;

- 1,562 heap pop/update observations; and
- exactly 340,150 Factor ticks in each solver.

After normalizing internal clause IDs and the implementations' distinct inprocessing-propagation accounting, the first nine decisions, first conflict order/set/levels, raw through final learned clause, retained counts, search choice, and top-32 order are also exact. All 76 Release tests pass. The tracked Kissat trace overlay cleanly rematerializes, configures, and compiles from the pinned source.

4 Decision-grade performance

The timing evidence is frozen in campaign directory

`benchmarks/runs/20260827-0680-tierd-rank4-factor-fresh-admission-timing`

It uses nine independent tasks, one solver invocation per task, on node018 at `-c 12 --mem=48000M`, with no `RLIMIT_AS`. All nine UNSAT proofs verify independently and residue is zero.

Arm	Median (s)	Spread	Conflicts
Candidate	10.733	0.4565%	647,074
Predecessor	15.340	0.6519%	983,053
Current Kissat	7.728	0.0129%	651,764

The candidate/predecessor median ratio is 0.699674, with paired range 0.695142–0.702868. The candidate/current-Kissat ratio is 1.388846, down from the predecessor's 1.984990. Candidate decisions are 942,227, versus 1,357,360 for the predecessor and 908,950 for Kissat. These state/work changes match the traced restored grid rather than measurement noise. Candidate Release binary SHA-256 is `f2eecac1...dcf`; timing summary SHA-256 is `ecdae42f...a55`.

5 Correctness gate and fuzz

The complete strict B28+C5 campaign is frozen in directory

`benchmarks/runs/20260827-0690-BC-rank4-factor-fresh-admission-guard`

Its array is 1652036 and aggregate is 1652102.

Gate measure	Predecessor	Candidate
Valid tasks	33/33	33/33
Checked SAT models	17	17
Verified UNSAT proofs	16	16
PAR-2 seconds	1,373.376	1,382.422
Maximum RSS (GiB)	2.844	2.843

The PAR-2 ratio is 1.006587, below the 1.10 guard. There is no solve loss, invalidity, cleanup failure, or certificate residue. Summary SHA-256 is `16a03cb2...845`.

Under the exclusive fuzz-ownership rule, Claude prepared, submitted, monitored, interpreted, and cleaned campaign `benchmarks/runs/20260827-0700-rank4-factor-fresh-admission-fuzz`. Claude's authoritative terminal verdict is PASS. It reports:

- 81/81 corpus replays and 30,000/30,000 generated cases, with zero failures;

- 15,967 independently checked original-CNF SAT models and 10,968 accepted native LRAT proofs;
- differential agreement on all 26,935 verdict cases;
- 50 physical cores, 100 logical CPUs, 400,000 MiB, and 50 workers on each generated leg, without an address-space limit;
- direct reach of fresh arming, heap push, both-polarity admission, and an accepted fresh-seeded grid, plus negative controls outside DefaultPolicy; and
- zero crash, sanitizer/invariant finding, rejected model/proof, unexplained disagreement, or certificate/scratch residue.

Two prep floor-calibration attempts were classified before corrected prep 1652105 and legs 1652106–1652109 completed. No solver finding occurred and no Claude repair workflow was triggered.

6 Next action

Commit and push this isolated increment, leaving the repository clean and synchronized. Then freeze the new accepted Tactix baseline and establish a three-wave paired Rank-32 baseline against the same current-Kissat comparator. The replacement Rank 1–40 screen measured Rank 32 at 416.933 versus 293.092 seconds; that single pair selects the witness but is not itself a decision-grade timing claim. Rank 32 will receive its own common-state localization, exact order/tick/scheduling comparison, and isolated evidence chain.